

Collision-Aware RES–ISS–CLF Control with QP-Based Tension Management for Human–Cable Interaction in Cable-Driven Parallel Robots

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Abstract—This paper presents a collision-aware control framework for cable-driven parallel robots (CDPRs) operating in shared workspaces where cables may wrap around human limbs. We derive the full six-degree-of-freedom (6-DOF) output dynamics of the CDPR platform via input–output feedback linearization, yielding a double-integrator error system for which a rapidly exponentially stabilizing input-to-state stable control Lyapunov function (RES–ISS–CLF) is systematically constructed using a continuous-time algebraic Riccati equation (CARE). The RES–ISS–CLF is embedded in a quadratic program (QP) that simultaneously enforces unilateral cable-tension constraints and a contact-triggered, time-varying tension upper bound that smoothly reduces the load on a colliding cable. Together, these provide exponential tracking convergence during nominal operation and input-to-state stability under bounded external contact wrenches. Simulation experiments on an 8-cable CDPR modeled after the CRAFT platform—using a fixed cylinder to emulate arm-like contact—show that the proposed RES–ISS–CLF–QP outperforms a PID–QP baseline and a nominal CLF–QP controller in trajectory tracking accuracy, post-contact oscillation suppression, and Lyapunov function decay rate. The framework provides the first formal stability guarantee for the human–cable collision scenario addressed in prior detection-and-management pipelines.

Index Terms—Cable-Driven Parallel Robots, physical human–robot interaction, collision management, control Lyapunov function, quadratic programming, input-to-state stability.

I. INTRODUCTION

Cable-Driven Parallel Robots (CDPRs) are parallel manipulators whose moving platform (MP) is actuated by unidirectional tension cables rather than rigid links. These characteristics confer high reconfigurability, large workspaces, and high payload-to-mass ratios, leading to deployments across industrial handling, entertainment, and construction [1], [2]. At the same time, cables can only pull, not push. They can sag, and their routing through space couples geometry, constraints, and dynamics. These features make closed-loop control and safety assurance non-trivial in human–robot collaboration (HRC) scenarios.

The high reconfigurability of CDPRs has hindered the establishment of standardized, off-the-shelf collaborative

components and procedures [3], [4]. Beyond this lack of standardization, modeling errors remain a first-order concern: geometric misplacement, pulley and routing simplifications, cable elongation and sag, and sensing biases all degrade pose estimation and tracking accuracy [5]–[7]. Dynamic uncertainties compound the problem. Nonlinear and rate-dependent cable stiffness and hysteresis, friction, and varying inertial properties are hard to identify and maintain across operating conditions [8], [9]. To cope, the literature explores adaptive and robust strategies (e.g., adaptive dynamics, sliding-mode variants) [10]–[12] and data-driven or optimization-based methods (e.g., deep RL, MPC) [13], [14]. Nevertheless, no single approach simultaneously ensures formal stability while enforcing unilateral cable constraints during contact-rich human–cable interactions.

In shared workspaces, CDPR cables may contact and wrap around human limbs or environmental objects, as shown in Fig. 1. Such cable–environment interference can create abrupt tension rises and impulsive loads if not detected and mitigated quickly [15]. Cable–cable interference further perturbs kinematics and can induce jerky motion or sudden releases [16]. Safe collaboration therefore demands controllers that enforce unilateral tension constraints, manage tensions proactively during collision and wrapping, and preserve tracking performance without exciting post-contact oscillations.

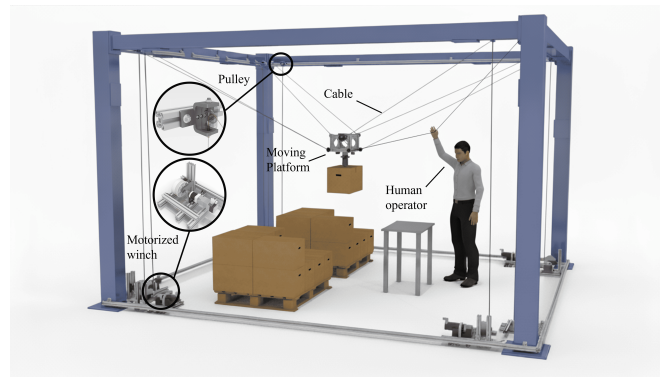


Fig. 1. Human–cable physical contact in the shared workspace of a CDPR.

Recent frameworks have introduced end-to-end pipelines for human–cable collision handling in CDPRs, including detection, cable identification, online management, and end-of-collision recovery [17]. Subsequent work extended these pipelines to incorporate cable-length and trajectory tracking

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during collisions, as well as broader carrying tasks involving multiple contact types [18], [19]. In these works, tension reduction during human–cable collision is achieved via PID-based corrections combined with sequential tension upper-bound schedules. While effective in practice, these strategies lack a closed-loop stability analysis and do not explicitly dissipate the excess energy that causes post-collision oscillations [19].

Control Lyapunov Function–Quadratic Programming (CLF–QP) approaches have been applied to CDPRs for principled trajectory-tracking design [20], and Lyapunov-based analyses exist for synchronous impedance control of MP–environment interactions [21]. To provide a clear comparison with these works, we highlight the following distinctions. Existing CLF–QP methods for CDPRs (e.g., [20]) consider nominal operation and platform-level disturbances; they do not model or detect human–cable wrapping events, do not apply a contact-triggered tension-reduction strategy, and do not provide input-to-state stability (ISS) guarantees under persistent external wrenches. ISS–CLF frameworks have been developed for generic nonlinear control-affine systems [22], [23], where the focus is on the mathematical conditions for ISS rather than on the CDPR-specific wrench geometry, unilateral cable constraints, or collision handling. The present work bridges these two lines: it derives the full 6-DOF CDPR output dynamics explicitly, constructs the RES–ISS–CLF via a systematic CARE procedure, and integrates a contact-triggered, time-varying tension upper bound into the QP constraint set – three elements that, to the best of our knowledge, have not been combined before for the human–cable collision scenario.

This paper closes this gap by presenting a collision-aware RES–ISS–CLF–QP control framework for CDPRs operating under human–cable interaction. The framework applies input–output feedback linearization to the full spatial (6-DOF) CDPR dynamics, derives the double-integrator error system explicitly, and constructs the RES–ISS–CLF via a CARE. A contact-triggered time-varying tension bound is integrated directly into the QP constraints, preserving both ISS guarantees and cable feasibility during and after collision. The system is validated in a wrapping simulator where a fixed cylinder emulates a human arm. The contributions of this paper are as follows:

- **C1 (CDPR dynamics modeling):** The full 6-DOF CDPR output dynamics are derived via input–output feedback linearization with the wrench-matrix geometry made explicit, extending the generic CLF–QP framework of [22], [23] to the cable-driven domain with unilateral constraints.
- **C2 (Stability under contact):** A RES–ISS–CLF controller provides the first formal exponential-ISS stability guarantee for the human–cable collision scenario, addressing the stability gap in the detection-and-management pipelines of [17]–[19], where PID-based tension reduction carried no closed-loop stability certificate.
- **C3 (Integrated tension management):** The contact-

triggered, time-varying tension upper bound from [18] is embedded as a hard QP constraint in the CLF–QP formulation, so that collision management and stability enforcement are unified in a single optimization.

- **C4 (Simulation evidence):** On an 8-cable wrapping scenario, the proposed RES–ISS–CLF–QP achieves tighter trajectory tracking, faster Lyapunov-function recovery, and lower post-contact oscillation energy than both a PID–QP baseline and a nominal CLF–QP, while remaining within cable-tension feasibility bounds.

Scope and assumptions. Full 6-DOF Cartesian control is considered. Cables are modeled as inelastic and massless. Contact is detected externally (e.g., from tension measurements), and the geometry is assumed known. All results are simulation-only; hardware validation on the CRAFT platform is identified as future work. Multi-body self-cable collisions are out of scope.

The remainder of this paper is organized as follows. Section II presents the system overview, collision/wrap simulator, and control architecture. Section III formulates the problem, derives the CDPR model, and develops the collision-aware RES–ISS–CLF–QP controller. Section IV reports simulation results and compares against PID–QP and CLF–QP baselines. Section V concludes and outlines directions for future work.

II. SYSTEM OVERVIEW AND SIMULATION SETUP

A. Control Architecture

Figure 2 depicts the overall control scheme. *Black* blocks form the nominal pipeline that commands cable tensions τ to realise the desired pose $\mathbf{p}_d$, orientation $\boldsymbol{\theta}_d$, twist $\mathbf{v}_d$, angular velocity $\boldsymbol{\omega}_d$, and their rates. *Red* blocks implement the collision-time tension-reduction policy layered on top of the nominal controller.

The collision management follows the sequential upper-bound reduction used in [17]. Upon detection of a collision at time t_{col} (identified via a tension-spike threshold) and identification of the involved cable c , a smooth time vector $s(t)$ with a 1 s transition is activated to avoid abrupt tension changes. Let $\mathbf{SI} \in \mathbb{R}^{m \times m}$ be a diagonal selector matrix that isolates the collided cable; the management matrix is

$$\mathbf{S}(t) = s(t) \cdot \mathbf{SI}.$$

This schedule contracts the time-varying upper bounds element-wise until the collided cable reaches its minimum allowable tension $\tau_{\min} \geq 0$. The bounds are fed to the tension-distribution (allocation) step, which solves [15], [17]–[19]

$$\begin{aligned} & \underset{\boldsymbol{\tau}}{\text{minimise}} \quad \|\boldsymbol{\tau}\|^2, \\ & \text{subject to} \quad \tau_{\min} \leq \boldsymbol{\tau} \leq (\mathbf{I} - \mathbf{S}(t))\boldsymbol{\tau}_{\max}. \end{aligned} \quad (1)$$

The proposed RES–ISS–CLF–QP block computes the stabilising platform command $\mathbf{u}^*$ (see (20)), and the allocator maps $\mathbf{u}^*$ to feasible tensions under the time-varying bounds.

Remark 2.1: The two disturbance types are treated separately. (i) *External contact wrenches* $\mathbf{w}_e$ arising from human–cable wrapping enter the dynamics as bounded, unknown input disturbances; the ISS bound in Definition 3.2 covers them

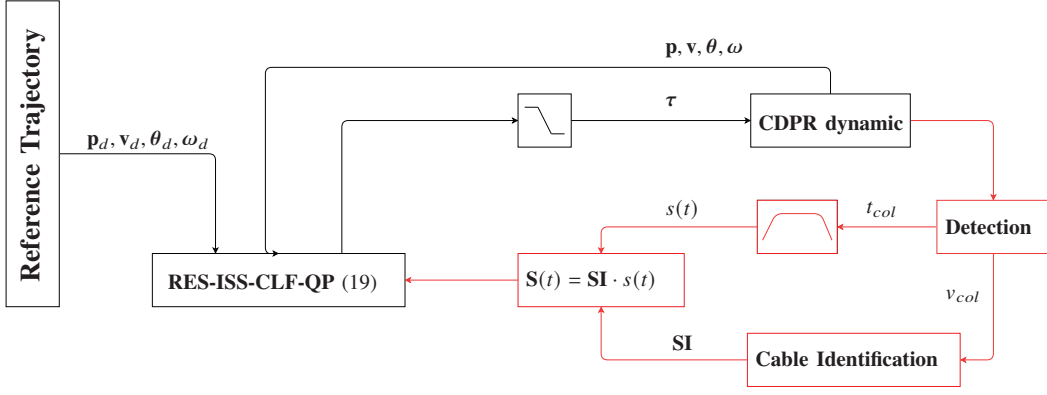


Fig. 2. Control scheme. Black: nominal RES-ISS-CLF-QP tracking with tension allocation. Red: collision management (detection, cable identification, and time-varying upper-bound reduction).

provided $\|\mathbf{d}\|_\infty < \bar{d}$ (Assumption 3.1). For short-duration wrapping events where cable tension limits physically bound the contact force, this assumption is mechanically reasonable. (ii) *Parametric uncertainty* in the platform mass m_g enters through $\mathbf{F}_p(\mathbf{x})$ and $\mathbf{G}_p(\mathbf{x})$ and is similarly captured by the lumped disturbance, but constitutes a structurally different perturbation that would require adaptive estimation for tight bounds. These two effects should not be conflated; the ISS framework guarantees ultimate boundedness for both, but does not adapt to parametric drift.

B. Collision/Wrap Simulator

The CRAFT platform at LS2N is an eight-cable suspended CDPR. Each actuation chain comprises a motor with integrated gearbox, a winch, and a cable end at the MP attachment point A_i (see Fig. 1). A simulator is developed using the same nominal parameters as the hardware, accounting for pulley geometry and full rigid-body CDPR dynamics. Control is executed in Cartesian space: inputs are tensions $\boldsymbol{\tau}$; outputs are platform pose $(\mathbf{p}, \boldsymbol{\theta})$, velocities $(\mathbf{v}, \boldsymbol{\omega})$, and cable lengths. The plant dynamics are integrated at 100 Hz and the controller (RES-ISS-CLF-QP and tension-allocation QP) runs at 50 Hz. Cables are modeled as inelastic and massless. Key simulation parameters are: $\varepsilon = 0.5$, $\bar{\varepsilon} = 1.7$, $\mathbf{Q} = 25\mathbf{I}$; the CARE solver is `care` and the QP solver is `quadprog` in MATLAB. The minimum cable tension is $\boldsymbol{\tau}_{\min} = \mathbf{0}\text{ N}$.

Modeling of Human-Cable Collision. Cable-human interaction (arm, neck, or leg) is represented as wrapping about a fixed cylinder located within the feasible workspace so that collided-cable tensions remain manageable. Upon contact, the cable route between the MP anchor B_i and the pulley exit A_i is split by two contact points C_i (near A_i) and D_i (near B_i , see Fig. 3). For quasi-static trajectories (no sliding) [16], the incoming and outgoing segments are tangent to the cylinder at C_i and D_i . Denoting by $\vec{\mathbf{t}}(\cdot)$ the unit tangent on the cylinder surface at arc-length parameters s_c and s_d ,

the geometric constraints read

$$\begin{aligned} \vec{\mathbf{t}}(s_c) &\parallel \overrightarrow{A_i C_i}, \\ \vec{\mathbf{t}}(s_d) &\parallel \overrightarrow{B_i D_i}. \end{aligned} \quad (2)$$

Enforcing (2) updates the effective cable lengths and directions, from which the simulator computes the cable displacement. The controllers (proposed and baseline) require only the *collision-onset* flag to trigger the time-varying tension caps and do not use detailed contact geometry, preserving method generality to other convex obstacles.

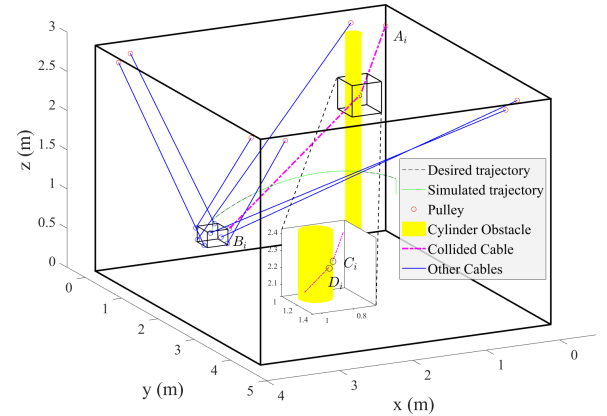


Fig. 3. Simulator with a fixed cylindrical obstacle. Cable i wraps around the cylinder, introducing two contact points C_i and D_i that split the route between the pulley exit A_i and the MP anchor B_i . In the quasi-static (no-slip) case, the entry/exit segments are tangent at C_i and D_i , enabling updated computation of cable displacement and lengths.

Reference trajectory. The reference motion is a three-segment pick-and-place path using minimum-jerk profiles between way-points, following [24], which yields smooth transitions and efficient point-to-point motion.

III. PROBLEM FORMULATION

This section defines the control problem, derives the full 6-DOF CDPR model, and develops the collision-aware RES-ISS-CLF-QP controller.

A. Preliminaries

Through the paper, $(\cdot)^\top$ denotes the transpose of a vector or matrix and $(\cdot)^{-1}$ denotes the matrix inverse. The output set is $\mathcal{Y} \subset \mathbb{R}^r$, the input set is $\mathcal{U} \subset \mathbb{R}^m$, and $\mathbb{R}_{>0}$ denotes strictly positive real numbers. Vectors are in bold lowercase (e.g., $\boldsymbol{\tau}$, $\boldsymbol{\eta}$) and matrices are in bold uppercase (e.g., $\mathbf{W}$, $\mathbf{P}$). Time dependence is omitted where not needed. The notation $\lambda_{\min}(\mathbf{A})$ and $\lambda_{\max}(\mathbf{A})$ denotes the minimum and maximum eigenvalues of a symmetric matrix $\mathbf{A}$, respectively.

B. Problem Definition

This paper addresses the control problem of guaranteeing the input-to-state stability of a CDPR under human–cable contact, where at least one cable wraps around a cylindrical obstacle. Formally, the goal is to design a control law that: (1) ensures exponential convergence of the full 6-DOF Cartesian tracking error to a neighbourhood of the origin during nominal operation, (2) maintains ultimate boundedness of the tracking error under bounded external contact wrenches, and (3) constrains all cable tensions within safe, time-varying bounds that mitigate collision loads without violating the cable unilateral constraint $\boldsymbol{\tau} \geq \mathbf{0}$.

The method addresses the input-to-state stability of the CDPR against external wrenches (human contact) and guarantees exponential convergence with ultimate boundedness. The QP layer enforces unilateral cable constraints and allocates tensions during collision and recovery.

C. Modeling of the Cable-Driven Parallel Robot

The dynamic model of a CDPR is formulated as a non-linear control-affine system in which cable tensions are the control inputs. Full spatial (6-DOF) rigid-body dynamics are used, capturing both translational and rotational motion. A CDPR is illustrated in Fig. 1, where the platform is actuated by m cables. The platform position is $\mathbf{p} = [x \ y \ z]^\top \in \mathbb{R}^3$ and orientation is $\boldsymbol{\theta} = [\phi \ \vartheta \ \psi]^\top \in \mathbb{R}^3$ (Euler angles, ZYX convention). The dynamic equation is

$$\mathbf{W}(\mathbf{p}, \boldsymbol{\theta})\boldsymbol{\tau} + \mathbf{w}_e + \mathbf{w}_g - \mathbf{I}_s(\boldsymbol{\theta})\dot{\mathbf{v}}_s - \mathbf{C}(\boldsymbol{\theta}, \boldsymbol{\omega}) = \mathbf{0}, \quad (3)$$

where $\boldsymbol{\tau} \in \mathbb{R}^m$ is the cable tension vector with $m = 8$, $\mathbf{W}(\mathbf{p}, \boldsymbol{\theta}) = [\mathbf{U}(\mathbf{p}, \boldsymbol{\theta})^\top \mathbf{M}(\mathbf{p}, \boldsymbol{\theta})^\top]^\top \in \mathbb{R}^{6 \times m}$ is the *wrench matrix* whose upper block $\mathbf{U} \in \mathbb{R}^{3 \times m}$ collects unit cable-force directions and whose lower block $\mathbf{M} \in \mathbb{R}^{3 \times m}$ collects the corresponding moment arms, $\mathbf{I}_s(\boldsymbol{\theta}) \in \mathbb{R}^{6 \times 6}$ is the spatial inertia matrix, $\mathbf{C}(\boldsymbol{\theta}, \boldsymbol{\omega}) \in \mathbb{R}^6$ is the Coriolis/centrifugal wrench, $\mathbf{w}_e \in \mathbb{R}^6$ is the external wrench (e.g., contact or field loads), $\mathbf{w}_g = [0 \ 0 \ -m_g g \ 0 \ 0 \ 0]^\top \in \mathbb{R}^6$ is the gravitational wrench (m_g is the platform mass, $g = 9.81 \text{ m s}^{-2}$) [25], and $\dot{\mathbf{v}}_s = [\ddot{\mathbf{p}}^\top \dot{\boldsymbol{\omega}}^\top]^\top$ is the spatial acceleration. The dynamics (3) are represented as a control-affine state-space system with state $\mathbf{x} := [\mathbf{p}^\top \ \boldsymbol{\theta}^\top \ \mathbf{v}^\top \ \boldsymbol{\omega}^\top]^\top \in \mathbb{R}^{12}$:

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} \mathbf{v} \\ T(\boldsymbol{\theta})\boldsymbol{\omega} \\ -\mathbf{I}_s^{-1}\mathbf{C}(\boldsymbol{\theta}, \boldsymbol{\omega}) \end{bmatrix}}_{f(\mathbf{x})} + \underbrace{\begin{bmatrix} \mathbf{0}_{3 \times m} \\ \mathbf{0}_{3 \times m} \\ \mathbf{I}_s^{-1}\mathbf{W}(\mathbf{p}, \boldsymbol{\theta}) \end{bmatrix}}_{g(\mathbf{x})} \boldsymbol{\tau} + \underbrace{\begin{bmatrix} \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{3 \times 1} \\ \mathbf{I}_s^{-1}\mathbf{w}_{eg} \end{bmatrix}}_{d(\mathbf{x})}, \quad (4)$$

where $\mathbf{v} = \dot{\mathbf{p}}$ is the translational velocity, $\boldsymbol{\omega}$ is the angular velocity in the body frame, $T(\boldsymbol{\theta})$ is the Euler-kinematics matrix (so that $\dot{\boldsymbol{\theta}} = T(\boldsymbol{\theta})\boldsymbol{\omega}$), and $\mathbf{w}_{eg} = \mathbf{w}_e + \mathbf{w}_g$ is the combined external and gravitational wrench. The term $d(\mathbf{x})$ represents the effect of external and gravitational wrenches on the platform acceleration.

Assumption 3.1 (Bounded disturbance): The disturbance d is an unknown function but is norm-bounded:

$$\|d\|_\infty := \sup_{t \geq 0} \|d(t)\| < \bar{d}, \quad (5)$$

for a known constant $\bar{d} > 0$. For short-duration wrapping contacts, the physical bound on cable tension limits $\mathbf{w}_e$, making this assumption mechanically reasonable.

The system output is defined as the full platform pose

$$\mathbf{y} = h(\mathbf{x}) = \begin{bmatrix} \mathbf{p} \\ \boldsymbol{\theta} \end{bmatrix} \in \mathbb{R}^6, \quad (6)$$

and the desired output trajectory is $\mathbf{y}_d(t) = [\mathbf{p}_d^\top \ \boldsymbol{\theta}_d^\top]^\top$. With the Euler kinematics $\dot{\boldsymbol{\theta}} = T(\boldsymbol{\theta})\boldsymbol{\omega}$, the first output derivative is $\dot{\mathbf{y}} = [\mathbf{v}^\top \ (T(\boldsymbol{\theta})\boldsymbol{\omega})^\top]^\top$, and the relative degree of the output with respect to the input $\boldsymbol{\tau}$ is determined by

$$\begin{aligned} L_g L_f^{i-1} h &= 0, \quad i = 1, \dots, r-1, \\ L_g L_f^{r-1} h &\neq 0, \end{aligned} \quad (7)$$

where $L_f h$ and $L_g h$ are the Lie derivatives of h along f and g , respectively. Since the translational control input $\boldsymbol{\tau}$ first appears in $\ddot{\mathbf{p}}$ (through $\mathbf{U}\boldsymbol{\tau}$) and in $\dot{\boldsymbol{\omega}}$ (through $\mathbf{M}\boldsymbol{\tau}$), the relative degree vector is $r = [2 \ 2 \ 2 \ 2 \ 2 \ 2]^\top$, i.e., $r = 2$ for each component of $\mathbf{y}$.

Definition 3.1: [23] A CLF on the output dynamics of (3) with $d \equiv 0$ is a continuously differentiable function V satisfying

$$c_1 \|\mathbf{x}\| \leq V(\mathbf{x}) \leq c_2 \|\mathbf{x}\|, \quad (8)$$

$$\inf_{\mathbf{u} \in \mathbb{R}^m} [L_f V(\mathbf{x}) + L_g V(\mathbf{x})\mathbf{u}] \leq -c_3 V(\mathbf{x}), \quad (9)$$

for positive scalars $c_1, c_2, c_3 \in \mathbb{R}_{>0}$. Condition (8) guarantees exponential stability of the nominal system. The corresponding CLF–QP controller is

$$\begin{aligned} \mathbf{u} &= \arg \min_{\mathbf{u} \in \mathbb{R}^m} \|\mathbf{u}\|^2 \\ \text{s.t.} \quad &L_f V(\mathbf{x}) + L_g V(\mathbf{x})\mathbf{u} \leq -c_3 V(\mathbf{x}) \\ &\mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max}. \end{aligned}$$

Under human–cable collision or sudden changes in platform mass, the CLF-controlled system may lose stability because the assumption $d \equiv 0$ is violated. An input-to-state stabilizing CLF (ISS–CLF) provides stability under these conditions.

Definition 3.2: [22] A continuously differentiable function V is a rapidly exponentially stabilising ISS–CLF if there exist positive scalars $c_1, c_2, c_3, \varepsilon \in (0, 1), \bar{\varepsilon}$ such that

$$c_1 \|\mathbf{x}\|^2 \leq V(\mathbf{x}) \leq \frac{c_2}{\varepsilon} \|\mathbf{x}\|^2 \quad (10)$$

$$L_f V(\mathbf{x}) + L_g V(\mathbf{x})\mathbf{u} \leq -\frac{c_3}{\varepsilon} V(\mathbf{x}) + \frac{1}{\bar{\varepsilon}} \|\mathbf{d}\|^2. \quad (11)$$

D. Designing the Collision-Aware RES-ISS-CLF-QP Controller

Finding a valid Lyapunov function for the nonlinear system (4) is generally challenging. The key insight is that feedback linearization of the output dynamics transforms the problem into a linear one for which a systematic CLF construction via LMI/CARE is available.

a) Step 1: Output choice, relative degree, and zero dynamics. The output $\mathbf{y} = [\mathbf{p}^\top \boldsymbol{\theta}^\top]^\top \in \mathbb{R}^6$ is chosen as the full platform pose because it directly encodes the control objective (6-DOF Cartesian trajectory tracking), is directly measurable from available sensors, and yields a uniform relative degree throughout the workspace.

Relative degree verification. From (4), neither $\dot{\mathbf{p}} = \mathbf{v}$ nor $\dot{\boldsymbol{\theta}} = T(\boldsymbol{\theta})\boldsymbol{\omega}$ contains the input $\boldsymbol{\tau}$, so $L_g h = 0$, i.e., the relative degree is at least 2 for every component of $\mathbf{y}$. Taking the second derivative: $\ddot{\mathbf{p}} = \mathbf{I}_{s,t}^{-1} \mathbf{U} \boldsymbol{\tau} + (\dots)$ and $\ddot{\boldsymbol{\theta}} = T(\boldsymbol{\theta}) \mathbf{I}_{s,r}^{-1} \mathbf{M} \boldsymbol{\tau} + (\dots)$, so the decoupling matrix $\mathbf{G}_p(\mathbf{x}) = L_g L_f h$ is non-zero wherever the wrench matrix $\mathbf{W} = [\mathbf{U}^\top \mathbf{M}^\top]^\top$ has full row rank (i.e., over the wrench-feasible workspace). Hence the relative degree vector is $\mathbf{r} = [2 \ 2 \ 2 \ 2 \ 2 \ 2]^\top$.

Absence of zero dynamics. The total relative degree is $r_{\text{tot}} = \sum r_i = 2 \times 6 = 12$, which equals the state-space dimension $n = 12$ (state $\mathbf{x} = [\mathbf{p}^\top \boldsymbol{\theta}^\top \mathbf{v}^\top \boldsymbol{\omega}^\top]^\top \in \mathbb{R}^{12}$). When $r_{\text{tot}} = n$, the input-output feedback linearisation is a global diffeomorphism on the state space, and no internal (zero-dynamics) subspace remains. Concretely, the coordinate change $(\boldsymbol{\eta}, \mathbf{z}) = \Phi(\mathbf{x})$ maps all 12 states to the 12-dimensional error-state $\boldsymbol{\eta} = [\mathbf{e}^\top \dot{\mathbf{e}}^\top]^\top$; the $\mathbf{z}$ -subsystem is empty. Stability of the closed-loop system therefore reduces entirely to stability of the canonical error dynamics (13), with no need to analyse internal dynamics separately [26]. Taking the second derivative of the output $\mathbf{y} = [\mathbf{p}^\top \boldsymbol{\theta}^\top]^\top$:

$$\ddot{\mathbf{y}} = \underbrace{\begin{bmatrix} \mathbf{0}_{3 \times 1} \\ \dot{T}(\boldsymbol{\theta}, \boldsymbol{\omega}) \boldsymbol{\omega} - T(\boldsymbol{\theta}) \mathbf{I}_s^{-1} (\boldsymbol{\omega} \times (\mathbf{I}_s \boldsymbol{\omega})) \end{bmatrix}}_{\mathbf{F}_p(\mathbf{x})} + \underbrace{\begin{bmatrix} \frac{1}{m_g} \mathbf{U}(\mathbf{p}, \boldsymbol{\theta}) \\ T(\boldsymbol{\theta}) \mathbf{I}_s^{-1} \mathbf{M}(\mathbf{p}, \boldsymbol{\theta}) \end{bmatrix}}_{\mathbf{G}_p(\mathbf{x})} \mathbf{u} + \underbrace{\begin{bmatrix} \frac{1}{m_g} \mathbf{F}_t \\ T(\boldsymbol{\theta}) \mathbf{I}_s^{-1} \mathbf{F}_r \end{bmatrix}}_{\mathbf{d}_{ij}(\mathbf{x})}, \quad (12)$$

where $\mathbf{u} = \boldsymbol{\tau}$ is the cable tension (control input), $\mathbf{U}(\mathbf{p}, \boldsymbol{\theta}) \in \mathbb{R}^{3 \times m}$ is the translational block of $\mathbf{W}$, $\mathbf{M}(\mathbf{p}, \boldsymbol{\theta}) \in \mathbb{R}^{3 \times m}$ is the rotational block, $\mathbf{F}_t = \mathbf{F}_e + \mathbf{F}_g \in \mathbb{R}^3$ is the translational component of $\mathbf{w}_{eg}$, $\mathbf{F}_r = -\boldsymbol{\omega} \times (\mathbf{I}_s \boldsymbol{\omega}) + \mathbf{T}_e + \mathbf{T}_g \in \mathbb{R}^3$ is the rotational component, and $\|\mathbf{d}_{ij}(t)\| \leq \bar{d}_{ij}$ for all $t \geq 0$. Note: the translational part of $\mathbf{F}_p$ is $\mathbf{0}_{3 \times 1}$ because, in the world frame, there is no translational Coriolis term for a rigid body—the drift in $\ddot{\mathbf{p}}$ is entirely captured by the disturbance $\mathbf{d}_{ij}$.

b) Step 2: Error-state canonical form. Define the output tracking error $\mathbf{e} := \mathbf{y} - \mathbf{y}_d = [\mathbf{p}^\top - \mathbf{p}_d^\top \boldsymbol{\theta}^\top - \boldsymbol{\theta}_d^\top]^\top \in \mathbb{R}^6$ and the augmented error state $\boldsymbol{\eta} := [\mathbf{e}^\top \dot{\mathbf{e}}^\top]^\top \in \mathbb{R}^{12}$. With the virtual control input $\mathbf{b} := \ddot{\mathbf{y}} - \ddot{\mathbf{y}}_d$ (the output-acceleration error from the feedback linearization), the error dynamics

take the double-integrator canonical form

$$\dot{\boldsymbol{\eta}} = \mathbf{F} \boldsymbol{\eta} + \mathbf{G} \mathbf{b}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{0}_{6 \times 6} & \mathbf{I}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \mathbf{0}_{6 \times 6} \\ \mathbf{I}_{6 \times 6} \end{bmatrix}, \quad (13)$$

i.e.,

$$\dot{\boldsymbol{\eta}} = \begin{bmatrix} \dot{\mathbf{e}} \\ \ddot{\mathbf{e}} \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{6 \times 6} & \mathbf{I}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} \end{bmatrix} \begin{bmatrix} \mathbf{e} \\ \dot{\mathbf{e}} \end{bmatrix} + \begin{bmatrix} \mathbf{0}_{6 \times 6} \\ \mathbf{I}_{6 \times 6} \end{bmatrix} \mathbf{b}. \quad (14)$$

The matrix $\mathbf{F}$ is not Hurwitz by itself; a stabilising feedback law must be assigned for $\mathbf{b}$.

c) Step 3: CARE-based CLF construction. To find a valid CLF for (13), we solve the Continuous-Time Algebraic Riccati Equation (CARE):

$$\mathbf{F}^\top \mathbf{P} + \mathbf{P} \mathbf{F} - \mathbf{P} \mathbf{G} \mathbf{G}^\top \mathbf{P} \preceq -\mathbf{Q}. \quad (15)$$

The CARE solution $\mathbf{P} = \mathbf{P}^\top \succ 0$ yields the *constant* gain matrix

$$\mathbf{K} = \frac{1}{2} \mathbf{G}^\top \mathbf{P} \in \mathbb{R}^{6 \times 12}, \quad (16)$$

and the linear state-feedback law $\mathbf{b} = -\mathbf{K} \boldsymbol{\eta}$. Assume that $\mathbf{K}$ is chosen so that the closed-loop matrix $\mathbf{A}_{cl} = \mathbf{F} - \mathbf{G} \mathbf{K}$ is Hurwitz. Then the quadratic function $V(\boldsymbol{\eta}) = \boldsymbol{\eta}^\top \mathbf{P} \boldsymbol{\eta}$ is a valid CLF satisfying

$$\lambda_{\min}(\mathbf{P}) \|\boldsymbol{\eta}\|^2 \leq V(\boldsymbol{\eta}) \leq \lambda_{\max}(\mathbf{P}) \|\boldsymbol{\eta}\|^2, \\ \inf_{\mathbf{b} \in \mathbb{R}^m} [L_{\mathbf{F}} V(\boldsymbol{\eta}) + L_{\mathbf{G}} V(\boldsymbol{\eta}) \mathbf{b}] \leq -\frac{\lambda_{\min}(\mathbf{Q})}{\lambda_{\max}(\mathbf{P})} V(\boldsymbol{\eta}). \quad (17)$$

The quadratic CLF $V(\boldsymbol{\eta}) = \boldsymbol{\eta}^\top \mathbf{P} \boldsymbol{\eta}$ is appropriate here because the error dynamics (13) form a linear double-integrator chain, for which a global quadratic CLF is guaranteed to exist by standard Lyapunov theory (the system is linear and stabilisable). The CARE-based design provides a solver-certified $\mathbf{P}$ directly and links the gain $\mathbf{K}$ to the ISS bound used in the next step.

Remark 3.1: The use of CARE rather than a simpler pole-placement method is motivated by the desire to obtain a $\mathbf{P}$ matrix that simultaneously satisfies both the CLF inequality (17) and the ISS bound in Definition 3.2, with an explicit relationship between the decay rate and the disturbance attenuation level. This joint construction is the mechanism linking the gain synthesis to the ISS guarantee.

d) Step 4: From virtual control to actual cable tensions. The virtual input $\mathbf{b} = -\mathbf{K} \boldsymbol{\eta}$ must be realised by the actual cable tensions $\mathbf{u} = \boldsymbol{\tau}$. From (12), in the nominal case ($\mathbf{d} \equiv \mathbf{0}$, consistent with Definition 3.2):

$$\mathbf{b} = \ddot{\mathbf{y}} - \ddot{\mathbf{y}}_d = \mathbf{F}_p(\mathbf{x}) + \mathbf{G}_p(\mathbf{x}) \mathbf{u} - \ddot{\mathbf{y}}_d. \quad (18)$$

Combining the CLF-ISS constraint with the tension bound constraints, the RES-ISS-CLF-QP controller is:

$$\mathbf{u}^* = \arg \min_{\mathbf{u} \in \mathbb{R}^m} \|\mathbf{F}_p(\mathbf{x}) + \mathbf{G}_p(\mathbf{x}) \mathbf{u} - \ddot{\mathbf{y}}_d\|^2 \quad (19) \\ \text{s.t. } \boldsymbol{\eta}^\top (\mathbf{F}^\top \mathbf{P} + \mathbf{P} \mathbf{F}) \boldsymbol{\eta} + 2 \boldsymbol{\eta}^\top \mathbf{P} \mathbf{G} (\mathbf{G}_p(\mathbf{x}) \mathbf{u} + \mathbf{F}_p(\mathbf{x}) - \ddot{\mathbf{y}}_d) \\ \leq -\frac{\lambda_{\min}(\mathbf{Q})}{\lambda_{\max}(\mathbf{P})} V_\epsilon(\boldsymbol{\eta}) + \frac{1}{\epsilon} \|\mathbf{P} \mathbf{G}\|^2 \\ \mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max},$$

where $V_\varepsilon(\eta) = \eta^\top \mathbf{P}_\varepsilon \eta$ and $\mathbf{P}_\varepsilon = \mathbf{P}/\varepsilon$. The CLF constraint in (19) enforces the ISS decay condition of Definition 3.2, ensuring that the Lyapunov function decreases at rate c_3/ε while tolerating bounded disturbances. When the slack variable of the CLF constraint is zero, the system is stable under (3) in the presence of bounded contact wrenches and parametric uncertainty.

e) Step 5: Integration with collision management.:

Equation (19) handles external wrenches on the *platform* (MP-level), but does not by itself limit tension in the individual cable contacting a human. Following [18], the time-varying upper bound from (1) is incorporated as a hard constraint, yielding the full collision-aware formulation:

$$\begin{aligned} \mathbf{u}^* &= \arg \min_{\mathbf{u} \in \mathbb{R}^m} \|\mathbf{F}_p(\mathbf{x}) + \mathbf{G}_p(\mathbf{x})\mathbf{u} - \ddot{\mathbf{y}}_d\|^2 \\ \text{s.t. } \quad & \eta^\top (\mathbf{F}^\top \mathbf{P} + \mathbf{P} \mathbf{F}) \eta + 2\eta^\top \mathbf{P} \mathbf{G} (\mathbf{G}_p(\mathbf{x})\mathbf{u} + \mathbf{F}_p(\mathbf{x}) - \ddot{\mathbf{y}}_d) \\ & \leq -\frac{\lambda_{\min}(\mathbf{Q})}{\lambda_{\max}(\mathbf{P})} V_\varepsilon(\eta) + \frac{1}{\varepsilon} \|2\eta^\top \mathbf{P} \mathbf{G}\|^2 \\ & \mathbf{u}_{\min} \leq \mathbf{u} \leq (\mathbf{I} - \mathbf{S}(t))\mathbf{u}_{\max}. \end{aligned} \quad (20)$$

Remark 3.2 (QP feasibility): Feasibility of (20) requires a non-empty constraint set, which holds whenever the wrench-feasible workspace is non-empty and the time-varying bound $(\mathbf{I} - \mathbf{S}(t))\mathbf{u}_{\max}$ remains above $\mathbf{u}_{\min} = \mathbf{0}$ for all t . The smooth schedule $\mathbf{S}(t)$ is designed to satisfy this throughout the simulated trajectory. A formal proof of persistent feasibility under general trajectories is identified as future work.

Remark 3.3 (Interaction between the time-varying tension bound and the ISS analysis): The time-varying upper bound $(\mathbf{I} - \mathbf{S}(t))\mathbf{u}_{\max}$ modifies the *constraint set* of the QP in (20), but does not enter the CLF construction itself. The RES-ISS-CLF $V(\eta) = \eta^\top \mathbf{P} \eta$ and the decay-rate/disturbance-attenuation parameters (c_3/ε , $1/\varepsilon$) are determined entirely by the CARE solution $\mathbf{P}$, which is computed once offline. At every time step, the QP in (20) seeks the cable tension $\mathbf{u}^*$ that simultaneously (i) minimises the tracking objective, (ii) satisfies the ISS-CLF decay constraint, and (iii) respects the current tension bounds. Provided the QP remains feasible (Remark 3.3), the ISS decay condition of Definition 3.2 is enforced at every step, and the ISS guarantee holds for all $t \geq 0$ regardless of how $\mathbf{S}(t)$ varies. In other words, the smooth temporal variation of the bounds does not invalidate the stability analysis: it only changes which tensions are available, not the Lyapunov certificate used to certify stability. Were the QP to become infeasible—a pathological case prevented by the smooth schedule—the ISS guarantee would need to be re-examined; this motivates the formal feasibility proof identified as future work.

IV. RESULTS AND ANALYSIS

This section presents simulation results evaluating the proposed framework for human–cable collision management. The scenario is a pick-and-place task on the 8-cable CDPR (Fig. 4) in which cable 2 wraps around a fixed cylinder at $t = 4.56$ s. Three controllers are compared:

- **RES-ISS-CLF-QP** (proposed): uses (20) with the collision-management schedule.

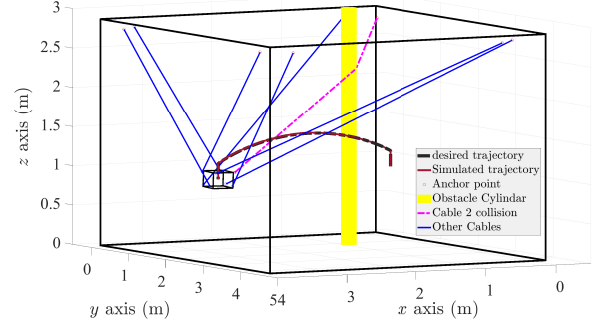


Fig. 4. Closed-loop CDPR simulation: 3-D pick-and-place task with human–cable collision management. The cylindrical obstacle is at the midpoint of the trajectory; cable 2 contacts the obstacle at $t = 4.56$ s.

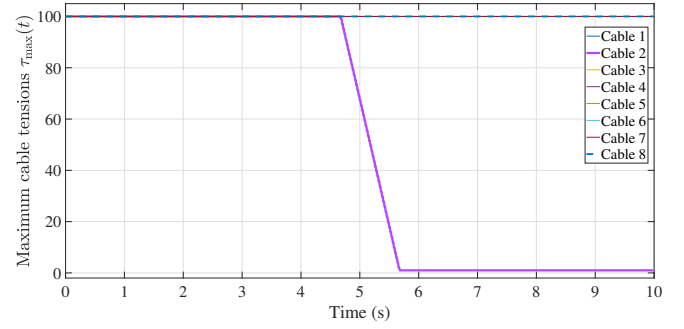


Fig. 5. Time-varying upper bound of cable 2 tension: $(\mathbf{I} - \mathbf{S}(t))\tau_{\max,2}$ (solid) versus time, with $\tau_{\min} = 0$ N. The reduction schedule activates at $t_{\text{col}} = 4.56$ s and transitions over 1 s.

- **CLF-QP**: uses the nominal CLF-QP (no ISS extension) with the same management schedule.
- **PID (no management)**: a standard PID controller without any tension bound reduction; included to illustrate the degradation when collision management is entirely absent.

All controllers use the same trajectory and QP allocator; only the stability constraint and disturbance handling differ.

Tracking performance (Fig. 6). All controllers *with* collision management (CLF-QP+man and RES-ISS-CLF-QP+man) maintain trajectory tracking through and after

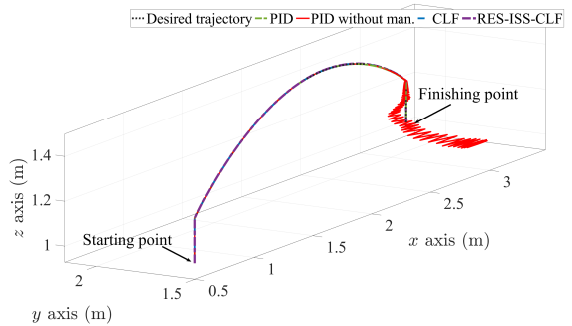


Fig. 6. Trajectory tracking performance of the three control laws under human–cable interaction at $t = 4.56$ s. *man* = with collision management schedule. Both CLF-QP and RES-ISS-CLF-QP with management maintain tracking; PID without management fails to recover.

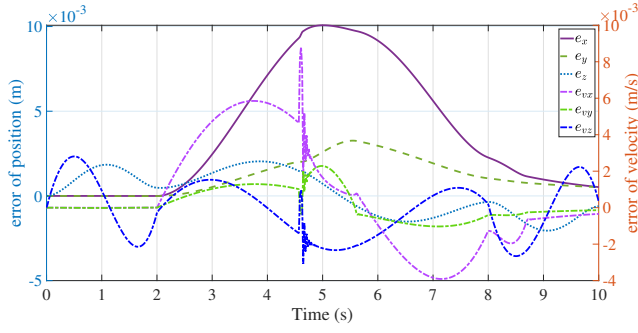


Fig. 7. Position and velocity tracking errors of the RES-ISS-CLF-QP controller with management versus time. Position errors ($\mathbf{e}_p$, left scale) and velocity errors ($\mathbf{e}_v$, right scale) are shown as separate line styles. The contact event at $t = 4.56$ s produces a transient that decays rapidly.

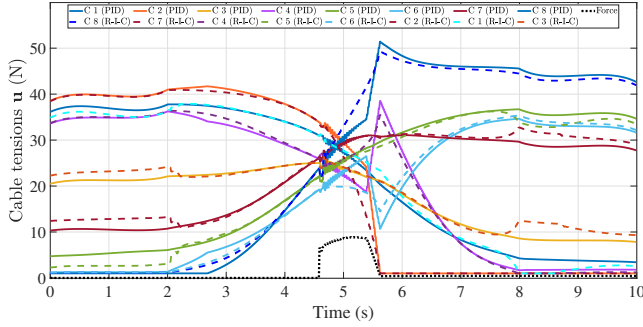


Fig. 8. Cable tensions of PID (top) and RES-ISS-CLF-QP (bottom) controlled systems. C = cable index, R-I-C = RES-ISS-CLF. Step-wise transitions visible near $t \approx 5.6$ – 5.7 s coincide with the QP solver switching feasible vertex at the boundary of the tension-reduction schedule; they remain within hardware limits and do not compromise stability (cf. Fig. 9).

the contact event. PID without management loses tracking entirely during the collision period, confirming that management is essential regardless of the controller type. The tracking trajectories of CLF-QP+man and RES-ISS-CLF-QP+man appear visually similar, because for the moderate disturbance level of a single-cable wrapping event, both controllers remain within the feasibility margin of the QP. The structural advantage of RES-ISS-CLF over the nominal CLF is therefore most visible in the Lyapunov function and error evolution, described below.

Stability evidence (Figs. 7, 9, 10). Figure 9 shows the Lyapunov functions of both CLF-QP and RES-ISS-CLF-QP under contact. The nominal CLF-QP cannot prevent a significant transient growth of V at the moment of contact; the RES-ISS-CLF-QP limits this growth and achieves a faster exponential recovery, consistent with the ISS decay rate c_3/ε guaranteed by Definition 3.2. Figure 10 shows $\dot{V}_{CLF}$ (the left-hand side of the stability constraint in (19)) remaining negative throughout the simulation for the proposed controller, including during and after contact—providing direct evidence that the ISS constraint is actively enforced. The values of $\dot{V}_{CLF}$ close to zero during nominal operation reflect the low steady-state tracking error.

Scope of the validation. These simulation results support contributions C2, C3, and C4 from the Introduction. They

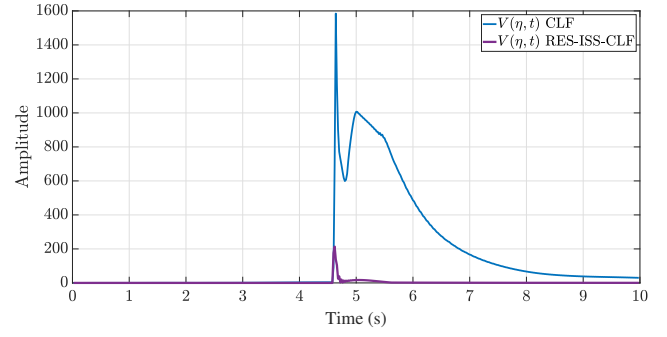


Fig. 9. Lyapunov functions $V(\boldsymbol{\eta})$ of CLF-QP and RES-ISS-CLF-QP versus time under human-cable interaction. The RES-ISS-CLF-QP limits the transient growth at contact and achieves faster exponential recovery.

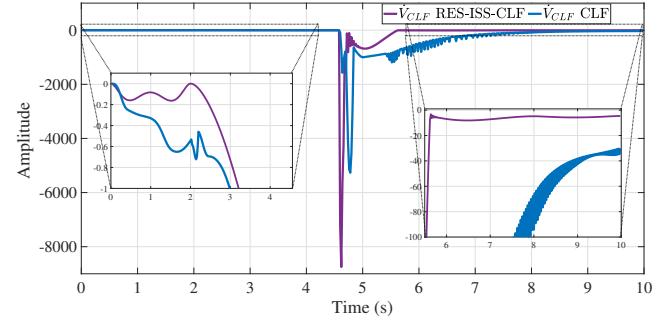


Fig. 10. Left-hand side of the CLF stability constraint $\dot{V}_{CLF}$ versus time for the proposed controller. Negative values throughout confirm that the ISS decay condition is actively satisfied, including during and after contact.

demonstrate that (i) the RES-ISS-CLF constraint provides a stronger stability guarantee than the nominal CLF under contact (C2), (ii) the integrated tension management preserves QP feasibility throughout the collision event (C3), and (iii) the proposed method outperforms PID without management in tracking recovery and outperforms nominal CLF in Lyapunov decay (C4).

The current evaluation is limited to *single-cable contact with a fixed cylindrical obstacle*. This scenario was chosen as a representative and reproducible baseline because: (i) single-cable wrapping is the most frequent human-cable collision mode in suspended CDPRs, where cable routing geometry makes simultaneous multi-cable contacts unlikely for typical task trajectories; (ii) a fixed cylinder is a conservative model for a stationary limb, producing the maximum wrap angle and contact force for a given configuration; and (iii) the fixed obstacle provides a fully controllable ground truth for Lyapunov and tension-profile analysis without confounding factors from obstacle motion. Extending to simultaneous multi-cable contacts and dynamic obstacles (e.g., a moving arm) is an important direction for future work. In multi-cable contact, multiple selector matrices $\mathbf{S}\mathbf{I}_c$ would be activated concurrently; the ISS analysis extends in principle since the disturbance term $\|\mathbf{d}\|_\infty$ bounds the combined wrench from all contacted cables, but the QP feasibility analysis becomes more involved as multiple tension bounds are tightened simultaneously.

The results also do not include sensor noise, actuator lag, or unmodelled cable dynamics. Hardware validation on the CRAFT platform is necessary before deployment claims can be made.

V. CONCLUSION

This work presented a RES-ISS-CLF-QP control framework for CDPRs operating under human-cable interaction. The design combines a feedback-linearization-based, CARE-synthesised rapid-exponential CLF for 6-DOF output tracking with an ISS bound for exogenous contact wrenches, and a QP-based tension allocator that enforces unilateral cable constraints. A contact-triggered, time-varying tension upper bound is integrated directly into the QP constraint set, unifying collision management and stability enforcement in a single optimisation. Using a wrapping simulator with a fixed cylindrical obstacle on an 8-cable CDPR model, the approach achieved tighter tracking, faster post-contact Lyapunov recovery, and lower oscillation energy than both a PID-QP baseline and a nominal CLF-QP, while maintaining QP feasibility throughout. The framework provides the first formal stability certificate for the human-cable collision scenario studied in prior detection-and-management pipelines.

The study assumed known geometry, inelastic massless cables, and frictionless wrap; all results are simulation-only. Identified limitations include: no formal QP feasibility proof under general trajectories, no hardware validation, and no treatment of multi-cable simultaneous contacts. Next steps include (i) hardware validation on the CRAFT platform with sensing noise and actuator lag; (ii) integration with compliance or trajectory reshaping to further dissipate post-contact energy; (iii) adaptive estimation of contact and disturbance bounds to tighten ISS margins; and (iv) extension to nonconvex obstacles and multi-cable contact.

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